SUPPLEMENTARY MATERIAL

Title: Cost-effectiveness of neoadjuvant pembrolizumab plus chemotherapy followed by adjuvant single-agent pembrolizumab for high-risk early-stage triple-negative breast cancer in the United States

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Table S1. One-way DSA and PSA parameters

Model parameter	Base-case	One-way DSA range	PSA distribution model	Notes
Female weight (kg)	69.65	95% CI limits estimated from the Normal distribution	Normal	-
Body surface area (m^2)	1.76	95% CI limits estimated from the Normal distribution	Normal	-
EFS - Pembrolizumab + chemotherapy / pembrolizumab	KM curve up to week 50 + Generalized Gamma distribution onwards	95% CI limits of the parameter estimates in the generalized gamma function	Random numbers generated from multivariate normal distribution model	Uncertainty in the parametric model is represented by the variance-covariance matrix of the parameter estimates.
EFS - Chemotherapy	KM curve up to week 50 + log normal distribution onwards	95% CI limits of the parameter estimates in the log normal function	Random numbers generated from multivariate normal distribution model	Uncertainty in the parametric model is represented by the variance-covariance matrix of the parameter estimates.
Exponential rate of LR to DM or death	0.0133	95% CI limits estimated from the normal distribution	Normal	-
% transition from LR to DM	0.9	95% CI limits estimated from the beta distribution	Beta	-
Exponential rate of DM to death - Pembrolizumab + chemotherapy / pembrolizumab	0.0192	95% CI limits estimated from the normal distribution	Normal	SE assumed to be equal to 20% of the base case value
Exponential rate of DM to death - Chemotherapy	0.0166	95% CI limits estimated from the normal distribution	Normal	SE assumed to be equal to 20% of the base case value
% LR	Based on KEYNOTE-522 trial data	95% CI limits estimated from the KEYNOTE-522 trial	Beta distributions using the mean and SE estimated from KEYNOTE-522	The mean and SE estimated from KEYNOTE-522
% DM	Based on KEYNOTE-522 trial data	95% CI limits estimated from the KEYNOTE-522 trial	Beta distributions using the mean and SE estimated from KEYNOTE-522	The mean and SE estimated from KEYNOTE-522
% received initial surgery	Based on KEYNOTE-522 trial data	95% CI limits estimated from the KEYNOTE-522 trial	Beta distributions using the mean and SE estimated from KEYNOTE-522	The mean and SE estimated from KEYNOTE-522

% received radiation	Based on KEYNOTE-522 trial data	95% CI limits estimated from the KEYNOTE-522 trial	Beta distributions using the mean and SE estimated from KEYNOTE-522	The mean and SE estimated from KEYNOTE-522
Utilities	Based on KEYNOTE-522 trial data	95% CI limits estimated from the KEYNOTE-522 trial	Beta distributions using the mean and SE estimated from KEYNOTE-522	The mean and SE estimated from KEYNOTE-522
Admin unit costs	Based on CMS Physician Fee Schedule (2021)	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Total metastatic treatment costs - Pembrolizumab + chemotherapy / pembrolizumab	59,467	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Total metastatic treatment costs - Chemotherapy	47,804	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Surgery cost	5,911	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Radiation cost	12,000	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Disease management costs	Based on Sieluk et al. (2022)	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
Terminal care costs	Based on Aly et al. (2019)	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
AE costs - Pembrolizumab + chemotherapy / pembrolizumab	1,931	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value
AE costs - Chemotherapy	1,547	95% CI limits estimated from the Gamma distribution	Gamma	SE assumed to be equal to 20% of the base-case value

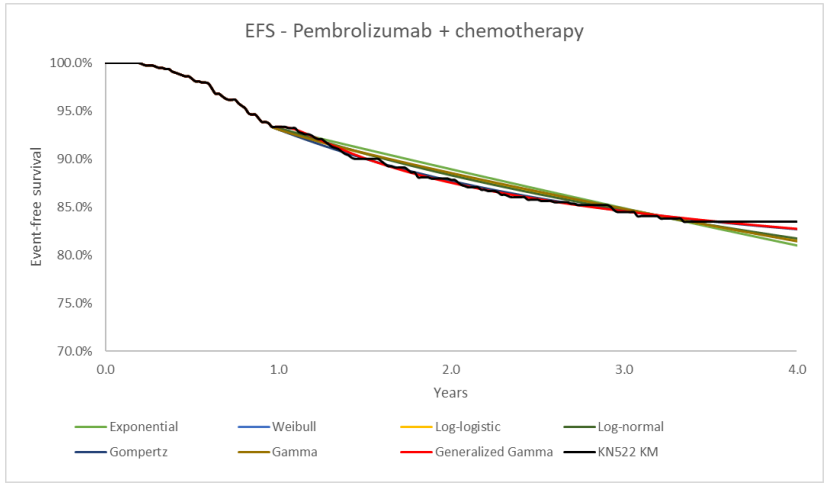
Abbreviations: AE, adverse event; CI, confidence interval; CMS, Centers for Medicare & Medicaid Services; EFS, event-free survival; DM, distant metastases; DSA, deterministic sensitivity analysis; KM, Kaplan-Meier; LR, locoregional recurrence; PSA, probabilistic sensitivity analysis; SE, standard error

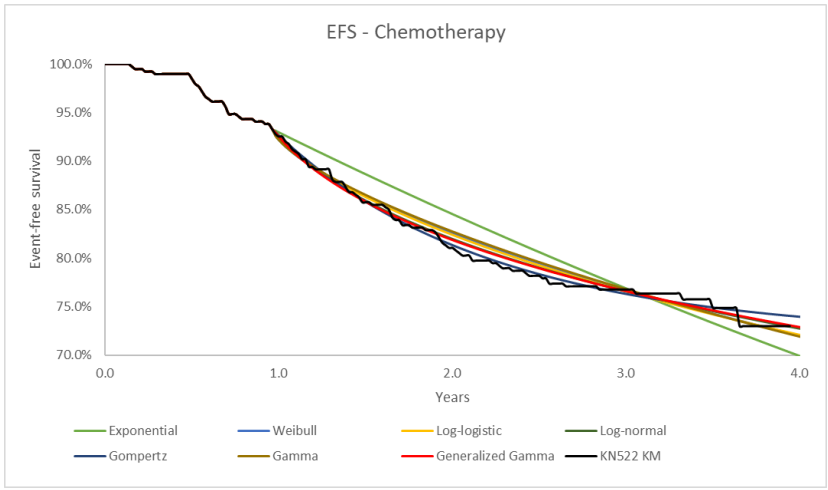
Table S2. Scenario analyses results

Scenarios	ICER (ΔCost/ΔQALY)
EFS	
EFS function - Pembrolizumab + chemotherapy / pembrolizumab - Piecewise - Week 50 - Log-normal	$57,569
EFS function - Pembrolizumab + chemotherapy / pembrolizumab - Piecewise - Week 68 - Log-normal	$34,664
EFS function - Pembrolizumab + chemotherapy / pembrolizumab - Piecewise - Week 43 - Log-normal	$44,526
EFS function - Chemotherapy - Piecewise - Week 50 - Generalized Gamma	$29,769
EFS function - Chemotherapy - Piecewise - Week 68 - Log-normal	$30,053
EFS function - Chemotherapy - Piecewise - Week 43 - Log-normal	$19,701
Utility	
Utility algorithm (US 5L crosswalk to 3L)	$27,384
Model specifications	
Time horizon (10 years)	$153,574
Time horizon (15 years)	$75,540
Time horizon (20 years)	$50,202
Annual discount rate - costs (0%)	$24,416
Annual discount rate - costs (5%)	$28,466
Annual discount rate - effectiveness (0%)	$14,883
Annual discount rate - effectiveness (5%)	$38,529
Remission after 8 years	$40,126
Treatment effect waning after 8 years	$45,273
Not applying relative dose intensity	$28,152
Consider life-time impact of chronic AEs	$28,328

All ICERs listed in 2021 United States dollars. Abbreviations: AE, adverse event; CI, confidence interval; DM, distant metastases; EFS, event-free survival; KM, Kaplan-Meier; ICER, incremental cost-effectiveness ratio; QALY, quality-adjusted life years; TOT, time on treatment.

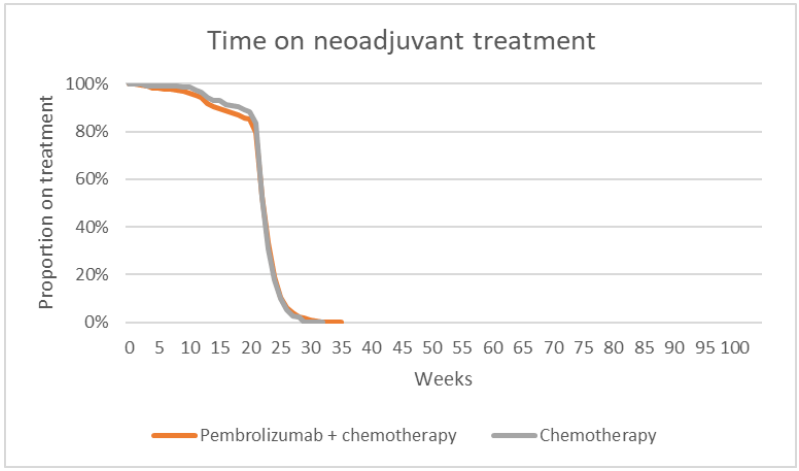
Figure S1. Extrapolations of event-free survival: Alternative parametric functions and base-case curves

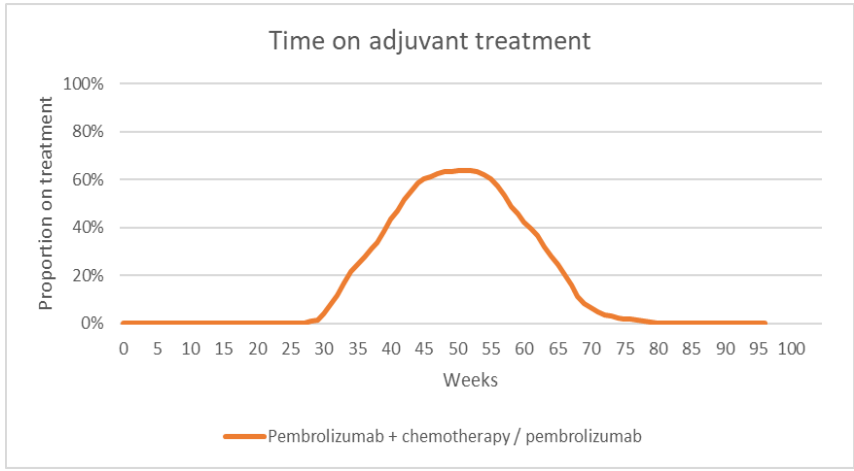




Abbreviations: EFS, event-free survival.

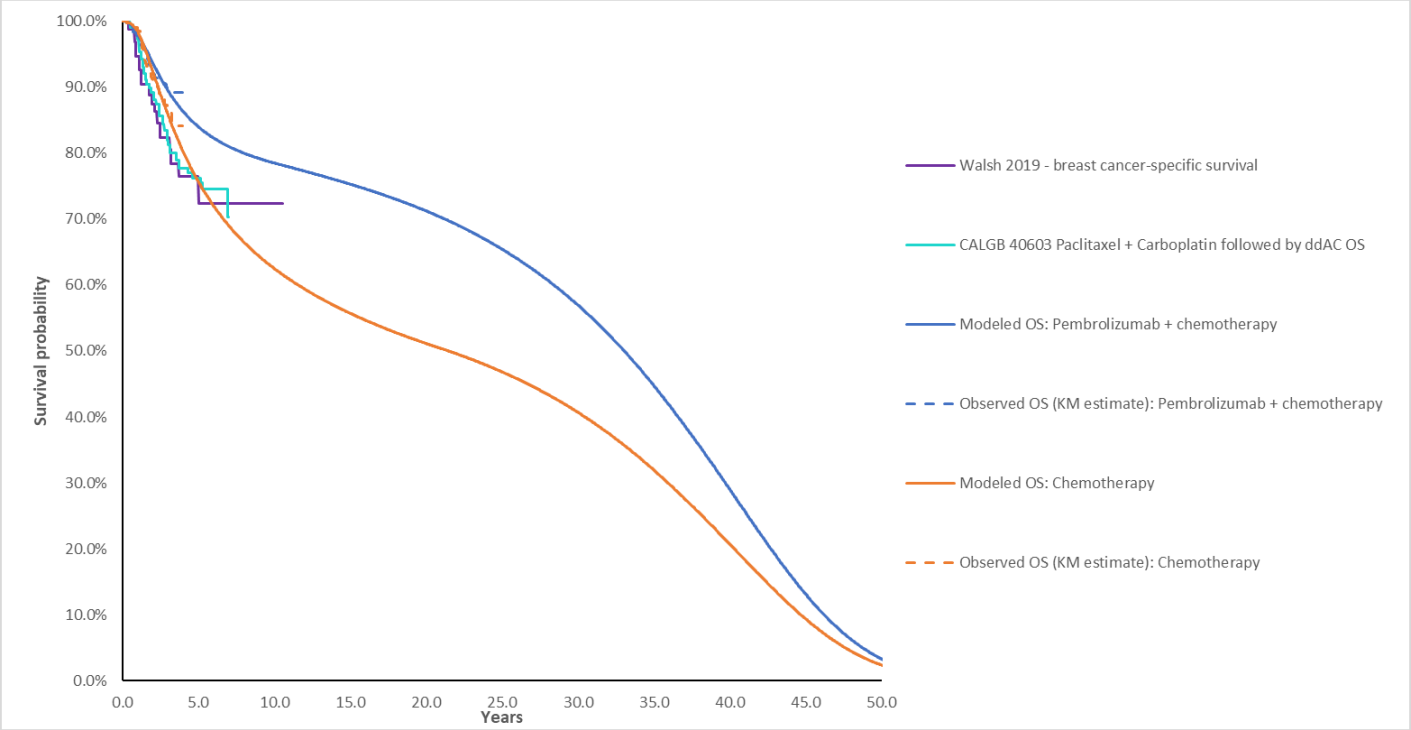
Figure S2. Time on treatment in neoadjuvant and adjuvant setting





Note: Patients in the chemotherapy arms received placebo as adjuvant treatment in the trial. So, the duration and costs of adjuvant treatment for these patients were not considered in the model.

Figure S3. Internal and external validation of modeled OS



Abbreviations: ddAC: dose-dense doxorubicin and cyclophosphamide; OS, overall survival.